

Let's Build a Map Quiz!

A Beginner Coding Story for Young Learners in India

Subject: Programming — JavaScript Fundamentals

Level: Beginner (Ages 10 – 14)

Duration: 60 – 90 minutes

Goal: Build an interactive India map quiz

What We Will Learn Together

By the time you finish reading this book, you will be able to explain what JavaScript is, create variables to store information, define functions to reuse code, write if-else statements to make smart decisions, respond to clicks, and assemble everything into a full working India map quiz!

What is JavaScript?

Before we start writing code, let us understand why we need JavaScript by looking at how a webpage is built.

HTML — The Shape

HTML gives every webpage its structure — headings, paragraphs, buttons, and images. Think of it as the outline of a beautiful Rangoli pattern drawn on the floor.

CSS — The Colours

CSS adds vibrant colors, fonts, and layout to the HTML structure. It fills the Rangoli outline with gorgeous hues.

JavaScript — The Diya that Lights It Up!

JavaScript makes pages respond to you! When you click a button, type in a box, or hover over an image, JavaScript runs your code and creates magic on screen. Without JavaScript, a webpage is like a printed painting — pretty, but still. With JavaScript, it comes alive!

You see JavaScript at work every day when search suggestions appear on Google, when score counters update in an online quiz, or when you track live buses on a map!

Our Three Building Blocks

Variables — The Dabba for Data

A variable is like a tiffin dabba (lunchbox). It stores a value and gives it a label so you can find it later. In JavaScript, we create a variable with the word `let`:

```
let correctState = 'Rajasthan';
```

Functions — School Experiment Steps

A function bundles a set of instructions so we can reuse them. Just like a science experiment has numbered steps that you follow every time, a function groups those steps under one friendly name:

```
function checkAnswer(clicked) {  
  // Our quiz-checking steps go here  
}
```

If-Else — The Cricket Umpire

An if-else statement works like a cricket umpire — it looks at what happened and gives a decision: OUT or NOT OUT!

```
if (clicked === correctState) {  
  alert('Sahi jawab!'); // Correct decision!  
} else {  
  alert('Galat!'); // Try again decision!  
}
```

Click Events: The Doorbell

Every time you click or tap on a screen, the browser creates an event. JavaScript is always listening — like a doorbell waiting for a visitor. We connect our buttons to our code using `onclick`:

```
<button onclick="checkAnswer('Rajasthan')">Rajasthan</button>
```

Putting The Complete Quiz Together

Now let us see how all four concepts combine into our finished interactive quiz:

```
// STEP 1: Variable (Dabba for data)  
let correctState = 'Rajasthan';  
let score = 0;  
  
// STEP 2: Function (Experiment Steps)  
function checkAnswer(clicked) {  
  // STEP 3: If-Else (Cricket Umpire Decision)  
  if (clicked === correctState) {  
    score = score + 10;  
  }  
}
```

```
    alert('Sahi jawab! +10 points! Score: ' + score);
  } else {
    alert('Galat! The correct answer was ' + correctState);
  }
}
```

Quick Reference Guide

Concept	Analogy	Description
Variable	Dabba	Stores a value with a label (let x = 5;)
Function	Experiment	Groups reusable steps together (function doIt() {})
If-Else	Umpire	Makes decisions based on conditions
Click Event	Doorbell	Runs code when user clicks (onclick)

Practice Challenge for Young Coders

Test Your Superpowers

Instructions: Read each story problem carefully and test your knowledge of variables, functions, decisions, and click events. Write your answers directly on the lines provided!

QUESTION 1

Which special word in JavaScript is used to create a new "Dabba" (variable) to store data?

- A) var_box
- B) let
- C) make
- D) store

QUESTION 2

Look at the code snippet below. What popup message will appear on screen if a student clicks on 'Kerala'?

```
let correctState = 'Gujarat';
function checkAnswer(clicked) {
  if (clicked === correctState) {
    alert('Win!');
  } else {
    alert('Try Again!');
  }
}
```

QUESTION 3

Match each JavaScript concept with its real-life analogy:

- | | |
|----------------------|---|
| 1. Variable | a) Doorbell ringing when pressed |
| 2. If-Else Statement | b) Tiffin Dabba holding food |
| 3. Function | c) Cricket Umpire making a decision |
| 4. Click Event | d) Science experiment steps followed together |

QUESTION 4

Fill in the blank to complete the click event so that clicking the button runs the function for 'Punjab':

```
<button _____="checkAnswer('Punjab')">Punjab</button>
```

QUESTION 5

Why is it helpful to group our code inside a **function** instead of writing the same if-else decision over and over again for every state button?
